

Yug Dabhi

yugdabhi974@gmail.com | <https://www.linkedin.com/in/yug-dabhi/> | <https://github.com/yugdabhi03/>

PROFESSIONAL SUMMARY

Software Developer specializing in secure full-stack applications and high-performance algorithms. Experienced in building tools that solve complex logistical and privacy challenges using modern web technologies and advanced data structures.

TECHNICAL SKILLS

Frontend Development: React.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS

Backend Development: Node.js, Express.js, Python, FastAPI, C/C++

Database Technologies: SQL, MongoDB, SQLite

Tools & Platforms: Git, AWS, Linux, RESTful APIs

PROJECTS

Route Optimizer System | React, Python, FastAPI, Algorithms

- Architected a full-stack navigation engine utilizing OSMnx graph data to generate efficient driving paths.
- Computed an N x N distance matrix via Dijkstra's algorithm for optimal and dynamic route calculation.
- Implemented an exact Traveling Salesperson Problem (TSP) solver for precise routing in small datasets.
- Engineered a Nearest-Neighbor & Node Relocation heuristic to optimize scale for large logistics operations.

Secure Notes Application | React, Node.js, Express, SQLite

- Developed a full-stack note-taking application focused on protecting sensitive data with robust security.
- Engineered server-side AES-256-GCM encryption to guarantee high-level data confidentiality at rest.
- Built a secure authentication and authorization pipeline utilizing JSON Web Tokens (JWT) and bcrypt.
- Designed a seamless, responsive React interface integrated with efficient RESTful APIs.

Orbit Task Manager | React, Node.js, Express, MongoDB, Tailwind

- Built a state-driven productivity application focusing on smooth micro-interactions and fast user experience.
- Engineered complex drag-and-drop task mechanics utilizing dnd-kit for intuitive task organization.
- Implemented dynamic state management and visual task analytics pipelines using Recharts.

Local File Search Engine | Python, Data Structures, Algorithms

- Developed a localized file indexing and search tool for optimized prefix and content-based queries.
- Engineered a custom Trie data structure achieving O(L) time complexity for rapid prefix searching.
- Built a persistent Inverted Index to rapidly retrieve keyword content across local file systems.

WORK EXPERIENCE

Amazon Delivery Station | Warehouse Associate

Sept 2025 - April 2026

- Collaborated in a fast-paced environment to ensure accurate processing and timely delivery of customer orders.
- Utilized warehouse management systems to track inventory and maintain high organizational standards.
- Adhered to strict safety protocols and performance metrics to consistently meet daily throughput goals.

Amazon Sortation Center | Warehouse Associate

Nov 2024 - Jan 2025

- Maintained high standards of operational efficiency while managing high-volume inventory flow.
- Sorted and staged outbound shipments to optimize delivery timelines and minimize errors.
- Resolved minor equipment and sorting issues to prevent bottlenecks in the sortation pipeline.

EDUCATION

Seneca Polytechnic | Computer Programming Diploma

May 2024 - Dec 2025

Program GPA: 3.2 | Relevant Coursework: Software Engineering, Data Structures, Algorithms, OOP.